



MODULE 1 • LAB • STUDENT PACKET

Introduction to Histology and the Microscope.

What histology is, how a compound light microscope actually works, how a slide gets made, and a practice set to work before you sit down at the bench.

Contents

1	What histology is	2
2	The compound light microscope	3
3	Magnification and resolution	5
4	Electron microscopes	6
5	How a slide gets made	7
6	Stains and what they tell you	8
7	Reading a slide well	9
8	Your first five minutes at the scope	10
9	Practice set	12

HOW TO USE THIS PACKET

Read a chapter, then work the matching part of the practice set with the packet open. This packet does not contain the answers. Answers live on the course web page, so that you have to commit to an answer before you find out whether you were right. That gap is where the learning happens.

BRING TO LAB

Bring this packet, completed in pen, to your first histology lab. You will use the bench walkthrough in chapter 8 at the scope, and the labeling exercise is the same figure that shows up on your first lab practical.

1 What histology is

Histology is the study of tissues. The word is built from histos, meaning tissue, and logos, meaning study of. A tissue is a group of cells that share a structure and do a job together, plus whatever material those cells put around themselves.

Every organ in your body is built from four tissue types and nothing else. Once you can recognize those four, you can walk into any organ on any slide and start making sense of it.

Epithelial tissue

Covers surfaces and lines cavities. Cells packed tightly together, almost no space between them, always sitting on a basement membrane with a free surface on one side.

Connective tissue

Supports, binds, and fills. Few cells, a lot of matrix. The matrix is what you identify it by.

Muscle tissue

Contracts. Long cells, oriented in a direction, because pulling only works along a line.

Nervous tissue

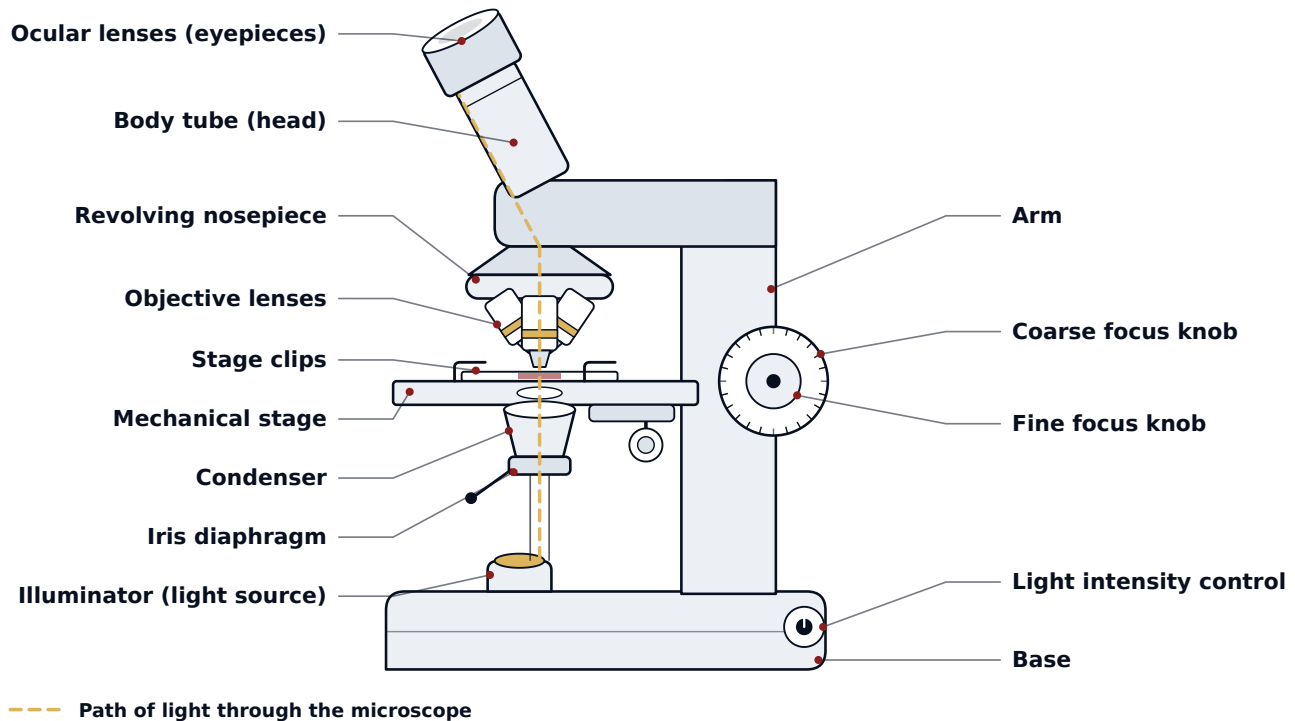
Detects and signals. Cells with processes reaching out to other cells.

WHY THIS MATTERS

Structure predicts function at every level, and histology is the level where that link is easiest to see. A cell built to absorb looks different from a cell built to protect. Once you get used to reading structure this way, the rest of the course gets easier, because you stop memorizing and start recognizing.

2 The compound light microscope

Compound means the image gets magnified twice, once by the objective lens and again by the ocular lens. Light comes up from underneath, passes through the specimen, and carries the image to your eye. That is why your slides have to be sliced thin enough for light to get through them.



The compound light microscope. The gold dashed line traces the light path: illuminator, iris diaphragm, condenser, specimen, objective lens, body tube, ocular lens, eye. Original figure drawn for BIO 004.

Mechanical parts

These are the parts that hold things and move things. They do not touch the image quality directly, but they are how you control what you are looking at.

Base and arm

The frame. The arm is where your hand goes when you carry it. Your other hand goes under the base.

Mechanical stage

The platform the slide sits on, with clips to hold it and two knobs that slide it left to right and front to back. Use the knobs. Do not shove the slide with your finger.

Coarse focus knob

The large knob. Moves the stage a long way fast. Scanning power only.

Fine focus knob

The small knob. Moves the stage a tiny amount. This is the only knob you touch on high power.

Revolving nosepiece

The turret the objectives are mounted in. Turn it by the ridged edge, not by grabbing an objective.

Optical parts

These are the parts that build and shape the image. If your image is bad, one of these is the reason.

Illuminator

The light source in the base, with an intensity control. Brightest is not best.

Iris diaphragm

An adjustable opening under the condenser. It controls how much light gets in, which controls contrast and depth of field. This is the most underused knob in the lab.

Condenser

A lens under the stage that gathers the light and concentrates it into a cone aimed at the specimen. Without it the light spreads out and the image goes soft.

Objective lenses

The lenses closest to the specimen. They make the first magnified image. Usually 4x scanning, 10x low power, 40x high power, and sometimes 100x oil immersion.

Ocular lenses

The eyepieces. Almost always 10x. They magnify the image the objective already made.

TWO WORDS WORTH KNOWING

Parfocal means that once you have focused on one objective, the image stays close to focus when you switch to the next one. You should only need fine focus after a switch. **Parcentric** means whatever was centered stays roughly centered. Both only work if you centered the specimen at low power first.

3 Magnification and resolution

These two get mixed up constantly, and they are not the same thing. Magnification is how much bigger. Resolution is how much detail.

Magnification

Total magnification is the objective power times the ocular power.

Standard lab scope, 10x oculars			
Objective	Name	Total magnification	What it is for
4x	Scanning	40x	Finding the specimen and reading the overall pattern. Always start here.
10x	Low power	100x	Identifying tissue type and layout.
40x	High power	400x	Cell shape, nuclei, cell junctions, fiber detail.
100x	Oil immersion	1000x	Blood cells and bacteria. Requires immersion oil.

There is a trade you make every time you go up in power. As magnification increases, the field of view shrinks, the image gets darker, and the depth of field gets shallower. That is why a specimen you never centered at 4x disappears entirely at 40x.

Resolution

Resolution is the smallest distance two points can be apart and still be seen as two points instead of one blur. Smaller number, better resolution.

A light microscope bottoms out at roughly 0.1 to 0.2 micrometers, because visible light has a wavelength and you cannot resolve detail much finer than the wavelength you are using to look at it. Past that point, more magnification just gives you a bigger blur. That is called empty magnification, and it is why a light microscope will never show you a ribosome no matter what number you put on the box.

SAY IT THIS WAY

Magnification makes it bigger. Resolution makes it clearer. Resolution is the one that limits you.

4 Electron microscopes

To get past the limit of visible light, you have to stop using light. Electron microscopes fire a beam of electrons at the specimen instead. Electrons behave with a far shorter wavelength, so the resolution improves by a factor of roughly fifty to one hundred, down to about 2 nanometers, at magnifications up to about 150,000x.

The two you need to tell apart		
	Transmission (TEM)	Scanning (SEM)
Beam does what	Passes through an ultrathin section	Sweeps across a coated surface
Image you get	Flat, two dimensional, like a very detailed slide	Three dimensional, like a photograph of a landscape
Shows you	The inside of a cell, organelle membranes, cristae	Surfaces and textures, microvilli, cilia, cell surfaces
Prep	Heavy metal stains, plastic embedding, sections around 100 nm	Specimen coated in a thin layer of gold

An easy check on an exam image: if it looks like a black and white slice with internal detail, it is TEM. If it looks like a moon landscape with shadows and depth, it is SEM.

5 How a slide gets made

Tissue does not come off a body ready to look at. It is wet, soft, colorless, and starting to break down. Every step below exists to solve one of those four problems, and knowing which problem a step solves is the fastest way to remember the order.

- ❶ **Fixation.** Chemically stops enzymes and microbes so the tissue quits degrading and holds its structure. Solves decay.
- ❷ **Dehydration.** Water is pulled out through a series of alcohol baths of increasing strength. Water and paraffin do not mix, so the water has to go before the wax can come in.
- ❸ **Clearing.** The alcohol is replaced with xylene, which mixes with both alcohol and paraffin and makes the tissue transparent. This is the bridge step.
- ❹ **Embedding.** The tissue is infiltrated with melted paraffin, which hardens. Solves softness. You cannot cut a thin slice of something floppy.
- ❺ **Sectioning.** The block is sliced on a microtome into ribbons 5 to 10 micrometers thick, thin enough for light to pass through.
- ❻ **Mounting.** Sections are floated onto a glass slide.
- ❼ **Staining.** Dyes are applied. Solves colorless. Then a coverslip goes on.

ORDER MNEMONIC

Fixation, **D**ehydration, **C**learing, **E**MBEDDING, **S**ectioning, **M**ounting, **S**taining. Fix Dead Cells Every Single Monday Sharp.

Electron microscopy runs the same sequence with harsher chemistry. Fixation uses glutaraldehyde and osmium tetroxide, the block is plastic instead of wax, and the sections come out near 100 nanometers. The stains are heavy metals such as uranyl acetate and lead citrate, because electrons are stopped by dense atoms rather than absorbed by colors.

6 Stains and what they tell you

A stain is not decoration. It is chemistry. Dyes bind to structures with a particular charge, so the color on the slide is telling you something about what that structure is made of.

Almost every slide you will see this term is stained with hematoxylin and eosin, written H and E. Hematoxylin is a basic dye that binds acidic structures, above all DNA, so nuclei come out blue to purple. Eosin is an acidic dye that binds basic structures, so cytoplasm, collagen, and most extracellular material come out pink.

Stains you may meet

Stain	Binds	Result
Hematoxylin	Acidic structures, mainly nucleic acids	Nuclei blue to purple
Eosin	Basic structures, proteins	Cytoplasm and collagen pink
Masson trichrome	Three dyes at once	Separates collagen from muscle by color. Good for connective tissue.
Wright or Giemsa	Blood cell components	Distinguishes white blood cell types on a smear
Silver stain	Reticular fibers, nerve processes	Black on a pale background
PAS	Carbohydrates, mucus, basement membrane	Magenta
Elastic stain	Elastic fibers	Dark wavy lines, useful in arteries

VOCABULARY

A structure that takes up basic dye and turns blue or purple is called **basophilic**, meaning base loving. A structure that takes up acidic dye and turns pink is called **acidophilic** or **eosinophilic**. If you can say why a nucleus is basophilic, you understand the whole system.

7 Reading a slide well

Most people lose points on lab practicals for reasons that have nothing to do with knowing the tissue. Here is what actually goes wrong.

Read the label, then do not trust it completely

The label tells you the organ. It does not tell you which part of the field the question is about. A slide labeled trachea contains epithelium, cartilage, glands, and connective tissue. Find the part being asked about.

Look at the slide before you look through the scope

Hold it up to the light for two seconds. You will see the outline, where the tissue is, and where the good section is, and you will save a minute of hunting.

Start low, every single time

At 40x total you can see the whole section and the overall architecture. At 400x you can see a hundred cells and no context. Architecture is usually what identifies a tissue, not the cells.

Do not overilluminate

A blazing bright field washes out exactly the faint edges you need. Close the iris diaphragm until the detail sharpens. Bright is not the same as clear.

Look at several examples

One slide of a tissue teaches you that slide. Three slides teach you the tissue. Textbook images are chosen because they are perfect. Your slide is a real section from a real block and may be torn, folded, or cut at an angle.

Think in three dimensions

You are looking at a two dimensional slice through a three dimensional object. Ask how the knife went through. A tube cut across gives you a ring. The same tube cut lengthwise gives you two parallel walls. Cut at an angle it gives you an oval that can look like something else entirely. Simple columnar epithelium cut obliquely can look stratified, and that mistake costs points every term.

8 Your first five minutes at the scope

Run this sequence every time until you stop having to think about it.

- 1 **Carry it correctly.** One hand under the base, one hand on the arm, held upright against your body. Oculars can fall out if you tip it. ☐
- 2 **Set up.** Plug in, turn the light on low, and clean the ocular and objective lenses with lens paper only. Paper towel scratches the coating permanently. ☐
- 3 **Start on scanning.** Click the 4x objective into place. You should feel and hear it seat. ☐
- 4 **Lower the stage all the way** with the coarse knob, then put the slide on with the coverslip facing up and secure it in the clips. ☐
- 5 **Raise the stage while watching from the side,** not through the oculars. Bring the objective close to the slide, then stop. ☐
- 6 **Now look through the oculars** and turn the coarse knob to lower the stage away until the image appears. Focusing away from the slide means you cannot drive the lens into it. ☐
- 7 **Adjust the oculars to your eyes.** Widen or narrow them until the two circles become one, then use the diopter ring so both eyes are sharp. ☐
- 8 **Center your specimen** with the mechanical stage knobs. Dead center, not near the edge. ☐
- 9 **Set the light.** Adjust intensity and close the iris diaphragm until edges look crispest. Do this before you go up in power, not after. ☐
- 10 **Click up to 10x, then 40x,** recentering each time. From 10x on, fine focus only. Never coarse focus on high power. ☐

SHUT DOWN

Rotate back to 4x, lower the stage, remove and clean your slide, turn the light down to zero before switching off, wipe any oil off the 100x lens, wrap the cord, and put the cover on. Leaving a scope on 40x with the stage up is how the next person breaks a slide.

TROUBLESHOOTING

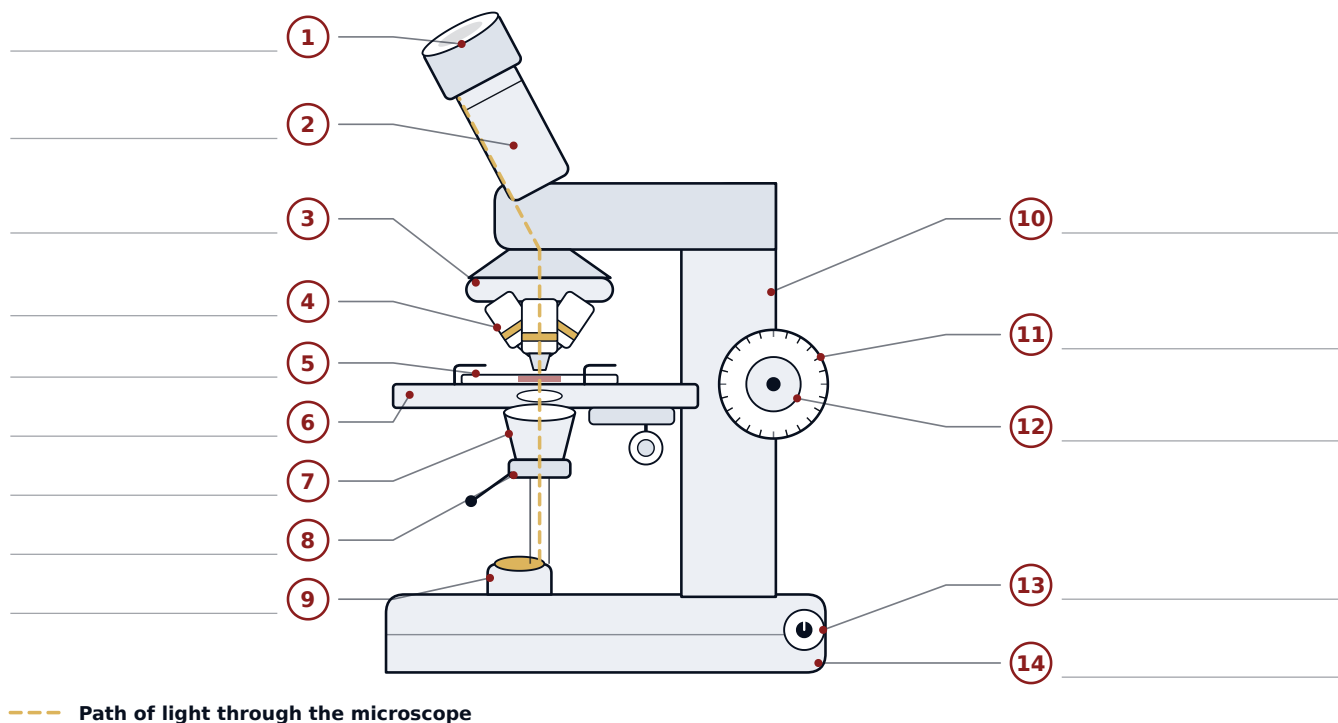
- **Half the field is dark.** The objective is not clicked fully into place, or the stage is not centered over the condenser. ☐
- **Specks that move when you rotate the ocular.** Dirt is on the ocular lens. ☐
- **Specks that stay put when you move the slide and the ocular.** Dirt is on the objective or the condenser. ☐
- **Image washed out and flat.** Too much light. Close the iris. ☐
- **Cannot find anything at 40x.** Go back to 4x, find it, center it, and climb again. ☐
- **Nothing focuses at all.** The slide is upside down. The coverslip must face the objective. ☐

9 Practice set

Work every item here before you open the answers. The answers live on the course web page, not in this packet, on purpose. Commit to an answer in pen first, then go check it. That is the part that makes it stick.

Part A. Label the microscope

Write the name of each numbered part on the line beside it.



Part B. What each part does

1. Which part concentrates the light into a cone aimed at the specimen?

- A. Ocular lens
- B. Condenser
- C. Iris diaphragm
- D. Objective lens

Answer: ____

2. Which part controls how much light reaches the specimen, and therefore the contrast and depth of field?

- A. Light intensity control
- B. Condenser
- C. Iris diaphragm
- D. Nosepiece

Answer: ____

3. Which lens produces the first magnified image of the specimen?

- A. Ocular lens
- B. Objective lens
- C. Condenser
- D. Iris diaphragm

Answer: ____

4. You need to move the slide slightly to the left without losing focus. What do you use?

5. Which focus knob is safe to use on the 40x objective, and why is the other one not?

6. Name the two parts you hold when you carry a microscope.

7. What does the revolving nosepiece do, and how should you turn it?

8. Your slide is in place but nothing at all comes into focus at any setting. What is the most likely cause?

Part C. Magnification math

9. Your oculars are 10x. What is the total magnification on the 4x, 10x, and 40x objectives?

10. With 10x oculars and a 100x oil immersion objective, what is the total magnification?

11. A scope gives 500x total with a 50x objective. What is the power of the ocular?

12. At 40x total magnification your field of view is 4.5 mm across. Roughly how wide is the field at 400x?

13. You count about 24 cells across the field at 100x total. Roughly how many will you see across the field at 400x?

14. Name three things that get worse as you increase magnification.

Part D. Resolution

15. What is the basic meaning of resolution?

- A. How many times larger the created image is than the real object
- B. How far apart two points can be and still be seen as separate
- C. How close together two points can be and still be seen as separate
- D. How bright the image is at a given magnification

Answer: ____

16. What is the approximate limit of resolution of a light microscope, and what sets that limit?

17. What is empty magnification?

18. True or false. A transmission electron microscope has both higher resolution and higher magnification than a light microscope.

Part E. Running the scope

19. You put a new slide on and go straight to the 40x objective. The field is empty. What went wrong and what do you do?

20. Your image is bright but flat, washed out, and edges are hard to see. What is the fix?

21. You switch from 10x to 40x and the image is nearly in focus already. What property is this, and which knob do you use now?

22. When you rotate from 10x to 40x, your specimen disappears off to the side. What does this tell you?

23. Half of your field is dark, in a clean curved edge. What are the two likeliest causes?

24. You see dark specks in the field. When you rotate the ocular, they rotate with it. Where is the dirt?

25. Why do you raise the stage while watching from the side rather than through the oculars?

26. List the shutdown steps for a lab scope in order.

Part F. Making the slide

27. Put the steps of slide preparation in order: clearing, dehydration, embedding, fixation, mounting, sectioning, staining.

28. Which step slows the deterioration of the specimen?

- A. Fixation
- B. Clearing
- C. Embedding
- D. Sectioning

Answer: ____

29. Why does the tissue have to be dehydrated before it can be embedded?

30. What does clearing with xylene accomplish?

31. Which tool is used to cut very thin sections of tissue?

- A. TEM
- B. SEM
- C. Condenser
- D. Dehydrator
- E. Microtome

Answer: ____

32. Why does a section have to be that thin?

33. Name two ways electron microscopy sample preparation differs from light microscopy preparation.

Part G. Stains

34. In an H and E preparation, which dye stains the nucleus and what color does it produce?

35. A structure on your slide is bright pink. Which dye bound to it, and what does that suggest about the structure?

36. Define basophilic and give one example of a basophilic structure.

37. Why is a stain necessary at all?

38. You need to tell collagen apart from smooth muscle in an artery wall. Which stain would you ask for and why?

Part H. Electron microscopy

39. Which electron microscope produces a two dimensional image of the inside of a cell, and how does it do it?

40. Which electron microscope produces a three dimensional view of a surface?

41. You want to see the cristae inside a mitochondrion. TEM or SEM?

42. You want to see the carpet of microvilli on the surface of an intestinal cell. TEM or SEM?

43. Why is an SEM specimen coated in gold?

Part I. Thinking in three dimensions

44. A blood vessel is cut across its short axis, and a second vessel is cut along its length. Describe how each one looks on the slide.

45. A slide of simple columnar epithelium appears to have several layers of nuclei at different heights. Give two possible explanations.

46. Why should you look at more than one slide of a given tissue before you decide you know it?

47. Before you look through the oculars, what is worth doing with the slide, and what does it buy you?

Now go check yourself.

You have committed to your answers. Open the course web page for this module and reveal the answers one item at a time.

Where the answers are

Course site, Module 1 Lab, Introduction to Histology and the Microscope. Scroll to section 9, Practice set, and use the Show answer button under each question.

What to do with the ones you missed

Do not just read the right answer and move on. Write, in your own words, what you thought the answer was and where your reasoning broke. Bring that note to lab.